CALCULATION REPORT - ANCHORAGE AND LAP LENGTHS

PJR-2026-003_WAREHOUSE (STEAM BOILER)

Document control

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National Annex	CEN_RECOMMENDED (Cambodia)
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Approved by	

Scope: this report covers ONE check only (EN 1992-1-1 bond/anchorage lengths) and reconciles it with sheet WH-STR-SD-004. It is not a complete structural design calculation. Report standard 10 is not yet authored; format is PROPOSED.

A. SCOPE

Anchorage and basic lap lengths for C30/37 concrete with fyk = 390 MPa are calculated to EN 1992-1-1 and compared with the table in section C of drawing WH-STR-SD-004 (R00).

B. DESIGN CODES AND NATIONAL ANNEX

Design codes (from design basis)

Standard	Generation	Edition	Used for
EN 1990	NOT RECORDED	NOT RECORDED	Basis of structural design
EN 1991-1-1	NOT RECORDED	NOT RECORDED	Self-weight, imposed loads
EN 1991-1-3	NOT RECORDED	NOT RECORDED	Snow
EN 1991-1-4	NOT RECORDED	NOT RECORDED	Wind
EN 1992-1-1	NOT RECORDED	NOT RECORDED	Concrete structures (RC)
EN 1993-1-1	NOT RECORDED	NOT RECORDED	Steel - general rules
EN 1993-1-3	NOT RECORDED	NOT RECORDED	Cold-formed members (purlins/girts)
EN 1993-1-5	NOT RECORDED	NOT RECORDED	Plated structural elements
EN 1993-1-8	NOT RECORDED	NOT RECORDED	Joints
EN 1997-1	NOT RECORDED	NOT RECORDED	Geotechnical design

Mixed 1st/2nd generation Eurocodes: User decision (P.Liheng, 14-OCT-2026): 1st and 2nd generation Eurocode parts may be combined on this project. The generation of each part must still be recorded per standard entry; compatibility of each combination is for the engineer to confirm.

National Annex: CEN_RECOMMENDED. Per SD-003: CEN recommended values used (no National Annex for Cambodia); local authority requirements apply where more onerous. Confirmed by user (P.Liheng) 14-OCT-2026.

C. MATERIALS

Materials (from design basis)

Id	Element	Specification	Status	Source
M-CONC-STUMP	Stump columns / stiffener columns	C30/37 XC1/XC4	confirmed	SD-004 A

Id	Element	Specification	Status	Source
M-CONC-PILE	Pile caps / tie beams / ground beams	C30/37 XC2	confirmed	SD-004 A
M-REBAR	Reinforcement	fyk = 390 MPa deformed bars, grade 390	confirmed	SD-004 A
M-STEEL-MAIN	Plates / welded sections / hot-rolled	S355JR (EN 10025-2)	confirmed	SD-005 A
M-BOLTS	Structural bolts	Class 8.8 / 10.9 (EN 14399-3 HRPV preloaded where noted)	confirmed	SD-005 A

D. ANCHORAGE AND LAP LENGTHS

CALC-ANCH-001 Anchorage and lap lengths, C30/37 with fyk = 390 MPa

Code reference: EN 1992-1-1 3.1.6, 8.4.2, 8.4.3

Inputs

Input	Value	Unit	Source
concrete class	C30/37	-	design basis M-CONC-PILE (SD-004 section A)
fyk	390.0	MPa	design basis M-REBAR (SD-004 section A)
gamma_c	1.5	-	EN 1992-1-1 Table 2.1N (CEN recommended)
gamma_s	1.15	-	EN 1992-1-1 Table 2.1N (CEN recommended)
alpha_ct	1.0	-	EN 1992-1-1 3.1.6(2)P (CEN recommended)
fctk,0.05	2.0	MPa	EN 1992-1-1 Table 3.1

Calculation

Step	Formula	Substitution	Result	Unit
Design tensile strength	fctd = alpha_ct * fctk,0.05 / gamma_c	1.0 * 2.0 / 1.5	1.333	MPa
Bond stress, good conditions	fbd = 2.25 * eta1 * eta2 * fctd (eta1=1.0, eta2=1.0)	2.25 * 1.0 * 1.0 * 1.333	3.0	MPa
Bond stress, poor conditions	fbd = 2.25 * eta1 * eta2 * fctd (eta1=0.7, eta2=1.0)	2.25 * 0.7 * 1.0 * 1.333	2.1	MPa
Design steel stress	sigma_sd = fyd = fyk / gamma_s	390 / 1.15	339.1	MPa

Results (unrounded; tabulated fctk,0.05)

Bar diameter (mm)	10	12	16	20	25
lb,rqd good bond (mm)	283	339	452	565	706
lb,rqd poor bond (mm)	404	484	646	808	1009
lap l0 good, 50% lapped (mm)	424	509	678	848	1060

*Scope limits: Basic required anchorage length only (alpha1-alpha5 = 1.0, lb,min not applied); lap = alpha6 * lb,rqd with alpha6 = 1.5. Straight bars.*

Reconciliation with SD-004 section C table (WH-STR-SD-004 as corrected: mandrel 20 mm = 140, 25 mm = 175)

Bar diameter (mm)	10	12	16	20	25
lbd good bond - drawing	280	335	445	560	700
lbd good bond - calc, drawing basis (fctk,0.05 = 2.03)	278	334	445	557	696
deviation (drawing - calc), mm	+2	+1	+0	+3	+4
lbd good bond - calc, Table 3.1 (fctk,0.05 = 2.0)	283	339	452	565	707
deviation (drawing - calc), mm	-3	-4	-7	-5	-7
lbd poor bond - drawing	400	480	640	800	1000
lbd poor bond - calc, drawing basis (fctk,0.05 = 2.03)	398	477	636	796	994
deviation (drawing - calc), mm	+2	+3	+4	+4	+6
lbd poor bond - calc, Table 3.1 (fctk,0.05 = 2.0)	404	484	646	807	1009
deviation (drawing - calc), mm	-4	-4	-6	-7	-9
lap l0 good (<=50% lapped) - drawing	420	505	670	840	1050
lap l0 good (<=50% lapped) - calc, drawing basis (fctk,0.05 = 2.03)	418	501	668	835	1044
deviation (drawing - calc), mm	+2	+4	+2	+5	+6
lap l0 good (<=50% lapped) - calc, Table 3.1 (fctk,0.05 = 2.0)	424	509	678	848	1060

Bar diameter (mm)	10	12	16	20	25
deviation (drawing - calc), mm	-4	-4	-8	-8	-10

*Drawing basis: fctk,0.05 = 0.7*fctm = 2.03 MPa -> fbd = 3.04 MPa.*

Drawing values agree with the calculation on the drawing's own basis within larger of 5 mm or 1 %: YES. Against the tabulated fctk,0.05 = 2.0 MPa the drawing values are shorter by up to 1.55 % (10 mm).

Verification status: CROSS-CHECKED against issued drawing; not independently reviewed by an engineer. Method: Reconciled with the SD-004 table (see reconciliation) and unit tests against hand values in tests/test_phase4.py.

E. CONCLUSIONS AND LIMITATIONS

The anchorage and lap lengths on SD-004 reproduce from the EN 1992-1-1 expressions on the drawing's own basis (fbd = 3.04 MPa), within the tolerance stated in the reconciliation. They are slightly shorter (about 1 to 1.6 %, see the reconciliation table) than values from the tabulated fctk,0.05 = 2.0 MPa; the engineer should decide which basis governs.

The mandrel-diameter row of SD-004 as issued (R00) gave 80 and 100 mm for 20 mm and 25 mm bars, which conflicts with EN 1992-1-1 8.3(1) and with the sheet's own note (7 phi above 16 mm = 140 and 175 mm). The corrected values are used in this calculation and have been applied to the sheet file; the engineer should confirm the sheet revision that carries the correction.

Workbook WH-STR-WB-001_R00.xlsx implements the same check with live formulas. Its results were tied out to this report's results file (CALC-ANCH-001) by tools/xlsxgen/verify_workbook.py, and it carries 8 hand-calculated checks.

Only basic required lengths are checked (alpha1-alpha5 = 1.0, no lb,min). The check was cross-checked against the issued drawing and unit tests; it has not been reviewed by an authorised engineer.

F. ASSUMPTIONS TO BE CONFIRMED

Id	Open item (from design basis)
F1	Foundation - precast square driven piles 300x300, L=7.0 m; soil investigation and pile load tests required
F2	Boiler / piping / crane / hoist and any platform loads not included
F3	Openings (doors, louvres, flue) not shown
F4	Terrain category II and no shielding, site altitude < 100 m
F5	Longitudinal (grid A/B) wind sway by braced bays - confirm
F6	Purlins/girts cold-formed - supplier verification for loads in SD-108
F7	Client name is a placeholder
F8	Eave sway limit h/100 = 120 mm - confirm with cladding supplier